



神雲科技股份有限公司

MiTAC Computing Technology Corporation

Document No. :

Document Name : TSR_HDD - HDD hot plug test for
Configurations with SAS Controller and StorageSpaces
incl. expander under HDD stress test

-----For configuration with Expander only!

Revision : 02

Document Title : TSR_HDD - HDD hot plug test for Configurations with SAS Controller and StorageSpaces incl. expander under HDD stress test -----For configuration with Expander only!	Doc. No. :	 神雲科技(股)公司
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版本變更 Revision History

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2024.05.29	01	First Released	Jason.ch.hsieh
2024.06.18	02	Format modification	Jacky Chiu

Prepared by	Checked by	Approved by
Jason.ch.hsieh	Jacky Chiu	

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1. 目的 Purpose

透過本文件了解如何測試 TSR_HDD 下列項目。

92	HDD Test	HDD hot plug test for Configurations with SAS Controller and StorageSpaces incl. expander under HDD stress test	Duration: 1x with HotSpare disk Verify HDD OOB Status in WebInterface before, while and after rebuild for correct information about HDD status. Requires cable from Mobo to HDD expander! Procedure for hot plug backplane: 1.) Create virtual disk with mirror or parity layout and hot spare disk	System running stable and OOB information is available and correct
		For configuration with Expander only!		

2. 範圍 Scope

本文件適用於測試 Fujitsu 伺服器產品人員及需要此文件內資訊的使用者。

3. 定義 Definition

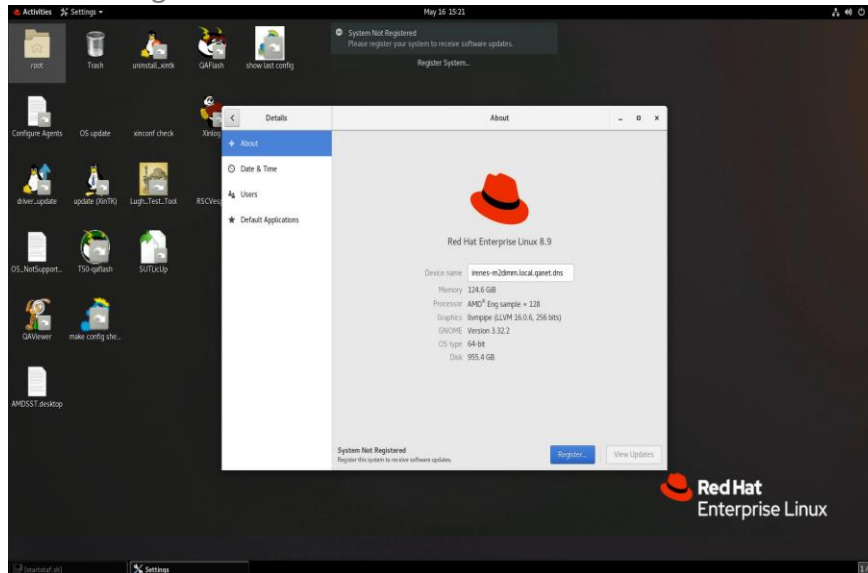
- 3.1 熱備源測試，最少需要將兩顆 HDD 建成一個 RAID(不可建為 RAID0)，且需留一顆 HDD 做熱備源，將建 RAID 的其中一顆 HDD 移除，做為熱備源的 HDD 會自動 rebuild。
- 3.2 建議做熱備源測試所使用的 HDD 使用同款相同容量的 HDD 進行測試。
- 3.3 測試過程確認 HDD 燈號及 GLED 燈號有無異常。
- 3.4 測試完畢後需要檢查 iRMC SEL 是否有異常並收集留存。

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4. 作業流程 Operation Flow

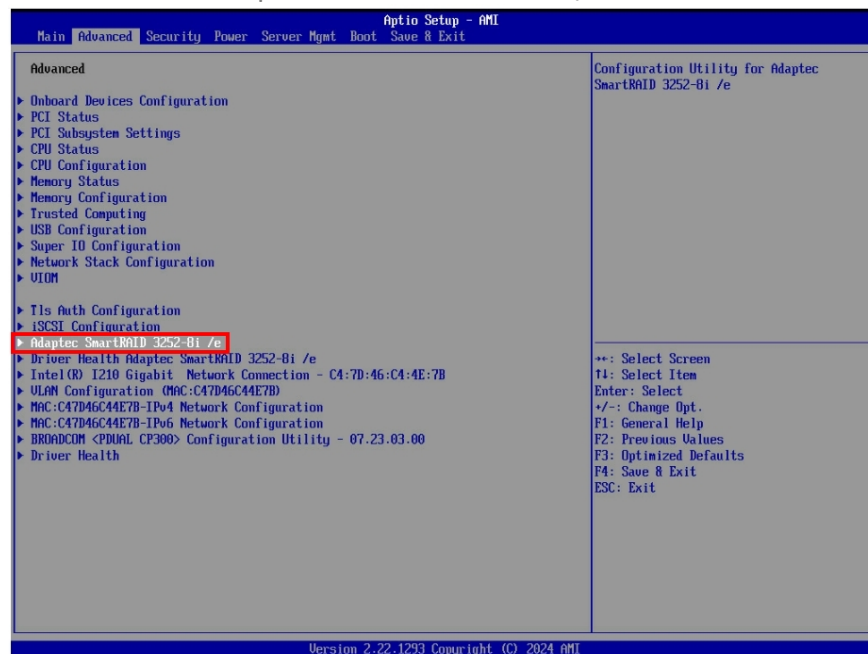
4.1 Setup Environment.

- >> Install RHEL 8 or 9 (Both SUSE or Windows are available)
- >> Install Xinconf by TiB
- >> Install Agnetless



4.2 BIOS Setup > Create a RAID for the hard drive

- >> Advanced > Adaptec SmartRAID 3252-8i /e



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>> Array Configuration > Create Array



Advanced Aptio Setup - AMI

▶ **Create Array**

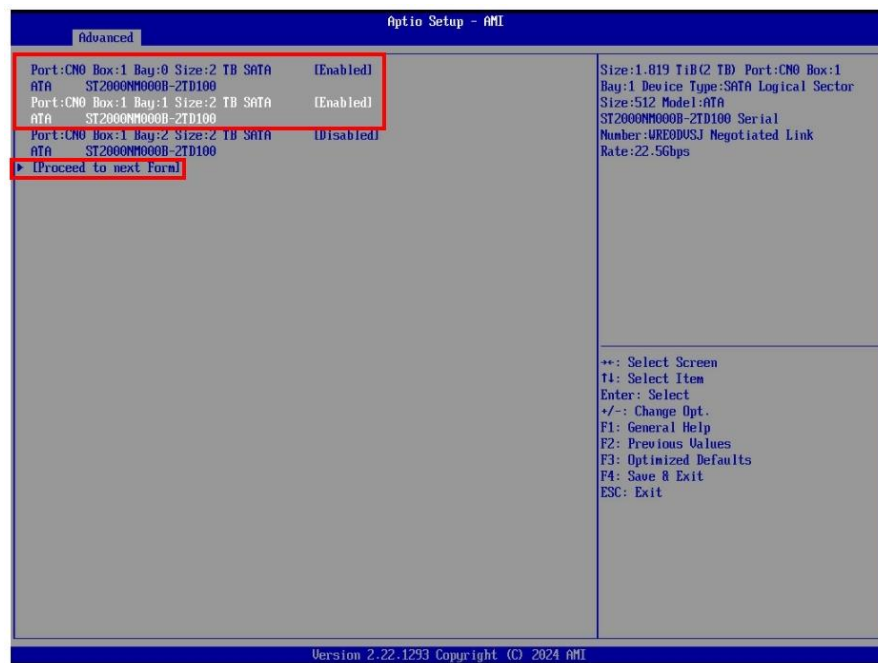
- ▶ Create Arrays with RAID0
- ▶ Create maxCache Array
- ▶ Manage Arrays

Creates an array from the group of selected physical drives of same type. Once an array is created, the unused space is available for creating logical drives. SAS-SAS/SATA-SATA combination is only allowed. Mixed combination like SAS-SATA is not allowed.

++: Select Screen
 T1: Select Item
 Enter: Select
 +/-: Change Opt.
 F1: General Help
 F2: Previous Values
 F3: Optimized Defaults
 F4: Save & Exit
 ESC: Exit

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>>Select "Enable" of the hard disk to build RAID



Advanced Aptio Setup - AMI

Port: C00 Box: 1 Bay: 0 Size: 2 TB SATA (Enabled)
ATA ST2000NM0008-2TD100

Port: C00 Box: 1 Bay: 1 Size: 2 TB SATA (Enabled)
ATA ST2000NM0008-2TD100

Port: C00 Box: 1 Bay: 2 Size: 2 TB SATA (Disabled)
ATA ST2000NM0008-2TD100

▶ [Proceed to next form]

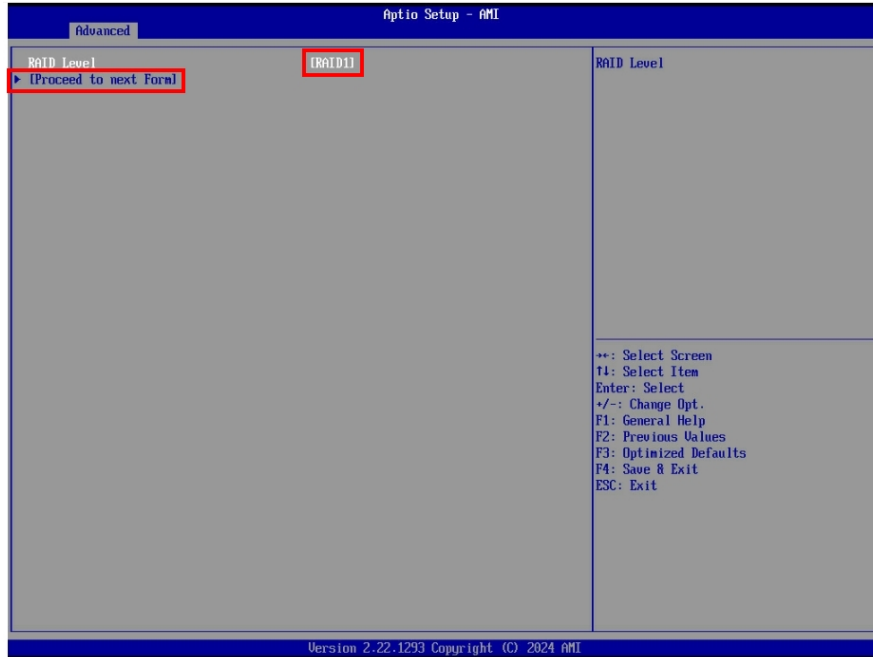
Size: 1.819 TiB (2 TB) Port: C00 Box: 1 Bay: 1 Device Type: SATA Logical Sector Size: 512 Model: ATA ST2000NM0008-2TD100 Serial Number: URE0DUSJ Negotiated Link Rate: 22.5Gbps

++: Select Screen
 T1: Select Item
 Enter: Select
 +/-: Change Opt.
 F1: General Help
 F2: Previous Values
 F3: Optimized Defaults
 F4: Save & Exit
 ESC: Exit

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Build RAID1



Advanced Optio Setup - AML

RAID Level [RAID1]

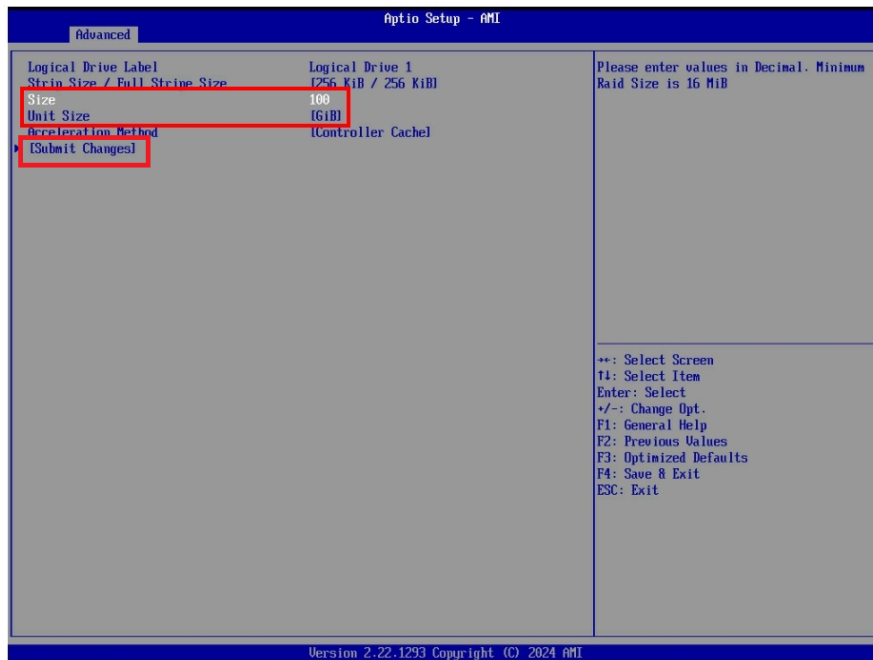
[Proceed to next Form]

RAID Level

** : Select Screen
 F1 : Select Item
 Enter : Select
 +/- : Change Opt.
 F1 : General Help
 F2 : Previous Values
 F3 : Optimized Defaults
 F4 : Save & Exit
 ESC : Exit

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After changing the capacity, please click > [Submit Changes]



Advanced Optio Setup - AML

Logical Drive Label Logical Drive 1

Strin Size / Full Strin Size (256 KiB / 256 KiB)

Size 100

Unit Size 1GiB

Acceleration Method (Controller Cache)

[Submit Changes]

Please enter values in Decimal. Minimum Raid Size is 16 MiB

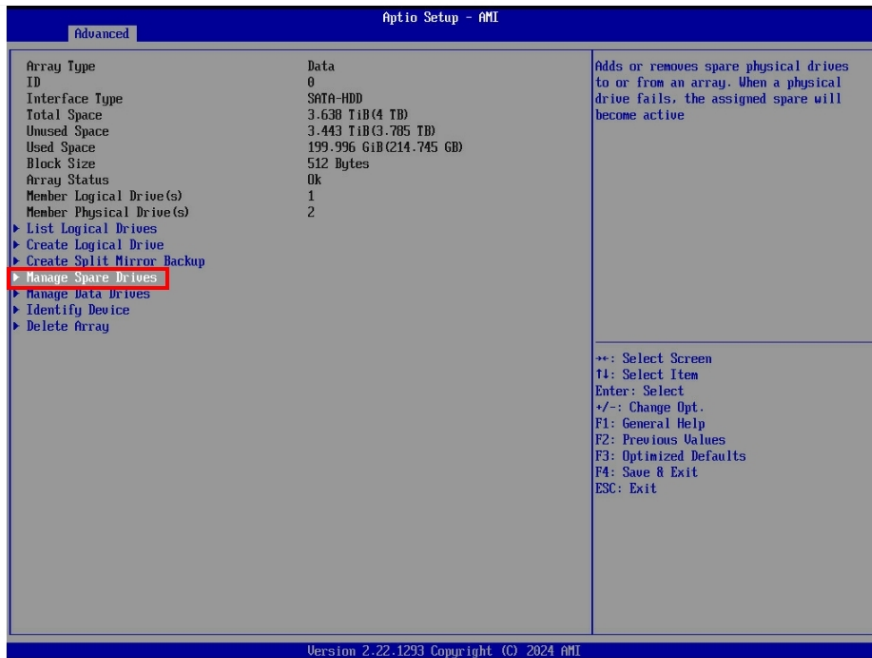
** : Select Screen
 F1 : Select Item
 Enter : Select
 +/- : Change Opt.
 F1 : General Help
 F2 : Previous Values
 F3 : Optimized Defaults
 F4 : Save & Exit
 ESC : Exit

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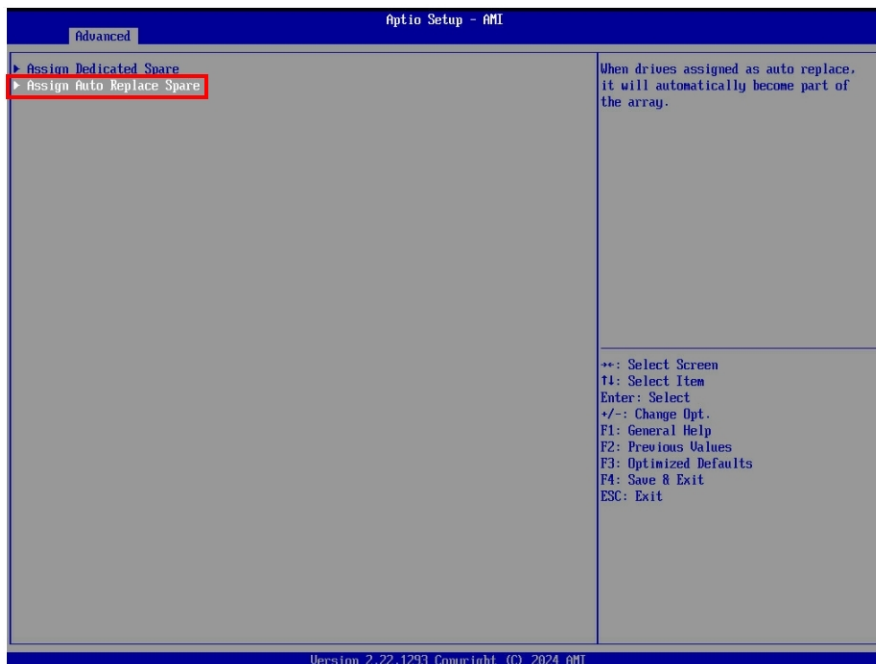
4.3 Set the idle hard drive as a hot spare source

Array Configuration > Manage Arrays > Array A > Manage Spare Drives



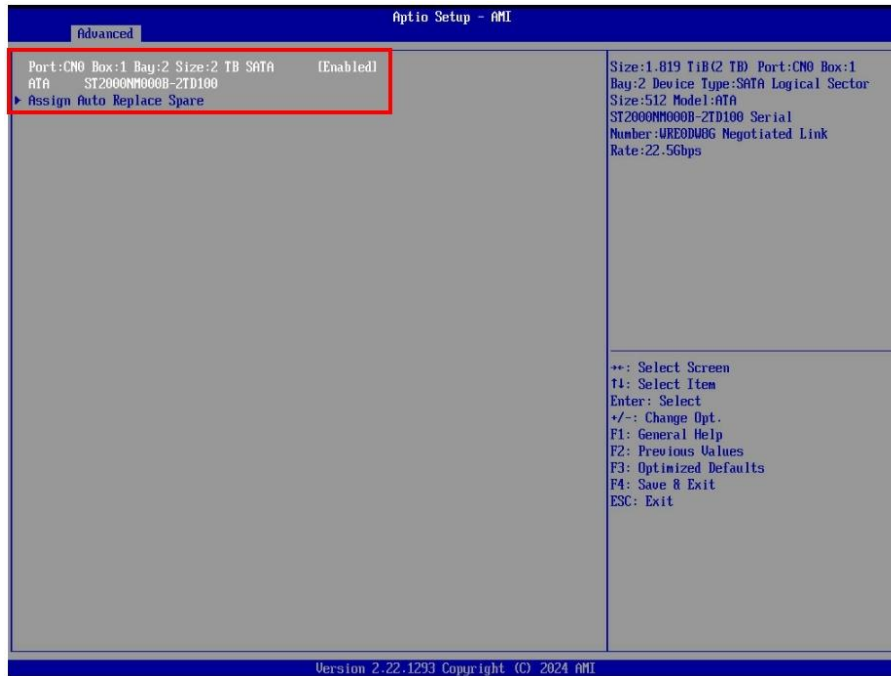
Select Assign Auto Replace Spare

*Select Auto to allow it to automatically detect failed hard drives and automatically rebuild them.



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Select the hard drive to be Hot Spare



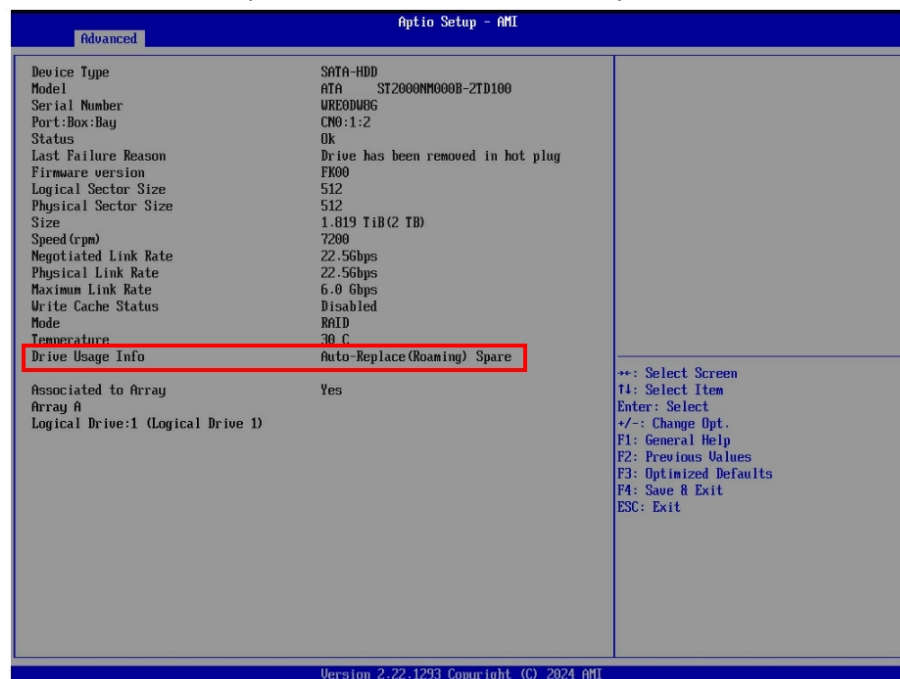
Advanced Aptio Setup - AMI

Port: C00 Box:1 Bay:2 Size:2 TB SATA (Enabled) ATA ST2000NM000B-2TD100 ▶ Assign Auto Replace Spare	Size:1.819 TiB(2 TB) Port:C00 Box:1 Bay:2 Device Type:SATA Logical Sector Size:512 Model:ATA ST2000NM000B-2TD100 Serial Number:URE0DU0G Negotiated Link Rate:22.5Gbps
--	--

** : Select Screen
 F1 : Select Item
 Enter : Select
 +/- : Change Opt.
 F1 : General Help
 F2 : Previous Values
 F3 : Optimized Defaults
 F4 : Save & Exit
 ESC : Exit

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Confirm that Hot Spare was created successfully



Advanced Aptio Setup - AMI

Device Type SATA-HDD Model ATA ST2000NM000B-2TD100 Serial Number URE0DU0G Port:Box:Bay C00:1:2 Status OK Last Failure Reason Drive has been removed in hot plug Firmware version F000 Logical Sector Size 512 Physical Sector Size 512 Size 1.819 TiB(2 TB) Speed (rpm) 7200 Negotiated Link Rate 22.5Gbps Physical Link Rate 22.5Gbps Maximum Link Rate 6.0 Gbps Write Cache Status Disabled Mode RAID Temperature 30 C Drive Usage Info Auto-Replace (Roaming) Spare	** : Select Screen F1 : Select Item Enter : Select +/- : Change Opt. F1 : General Help F2 : Previous Values F3 : Optimized Defaults F4 : Save & Exit ESC : Exit
--	---

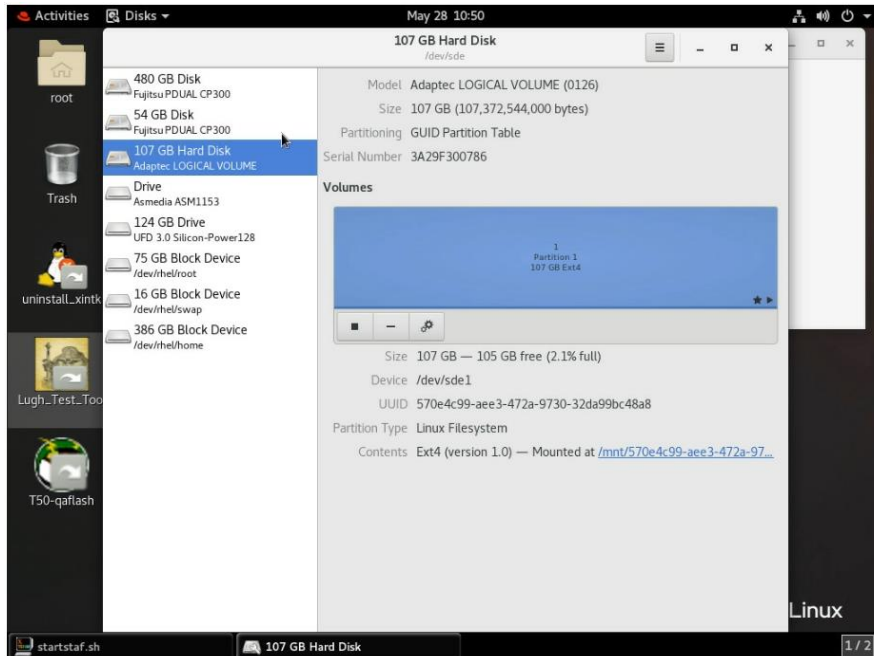
Associated to Array Yes
 Array A
 Logical Drive:1 (Logical Drive 1)

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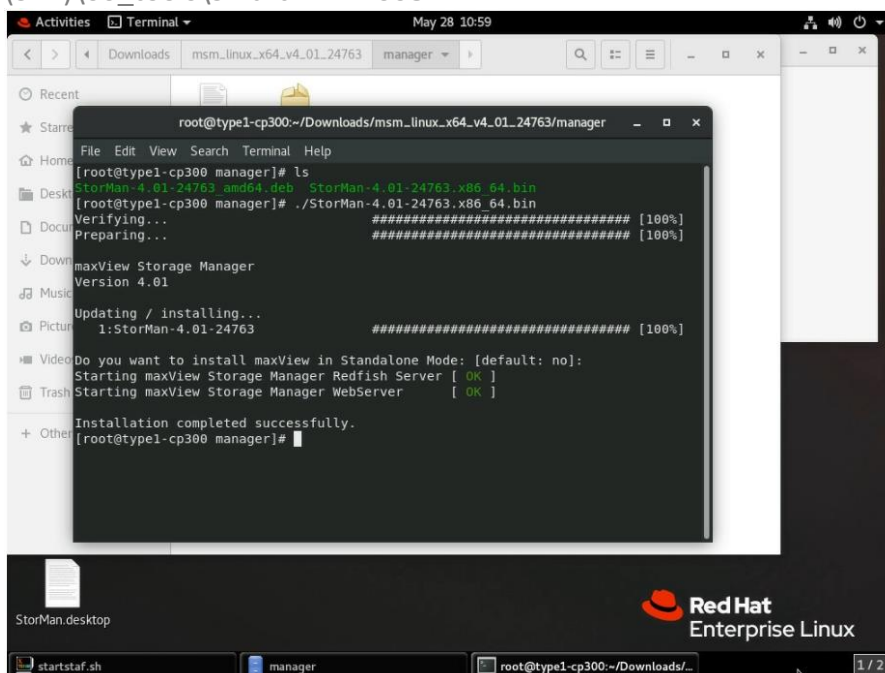
4.4 OS settings

Mount the hard drive you just created the RAID on



Install MaxView

Tool location: \\10.99.239.76\cdad\project\FJ\FJ_RX2530M7\Sapphire Rapids (SPR)\06_tools\SmartHBA21008i

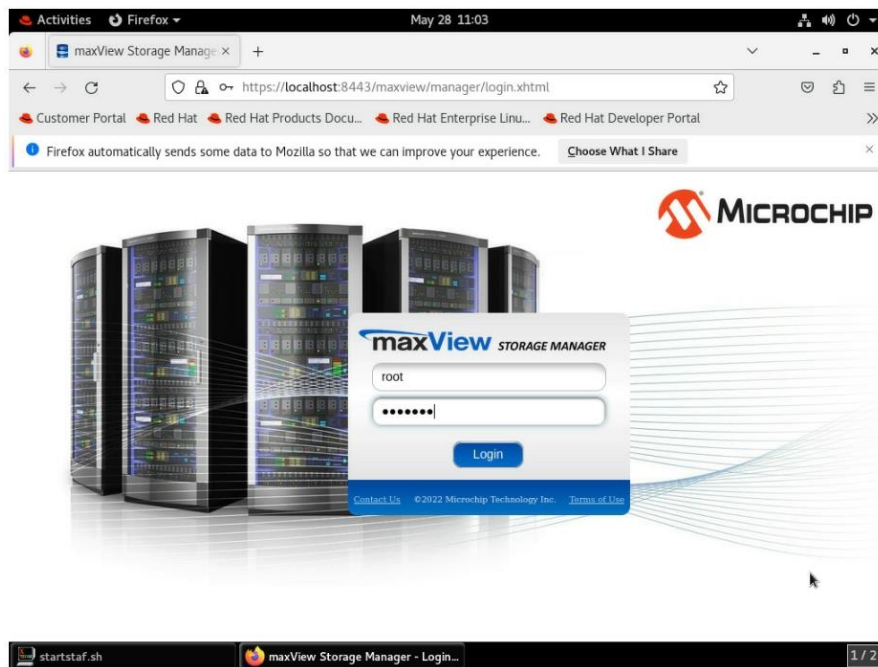


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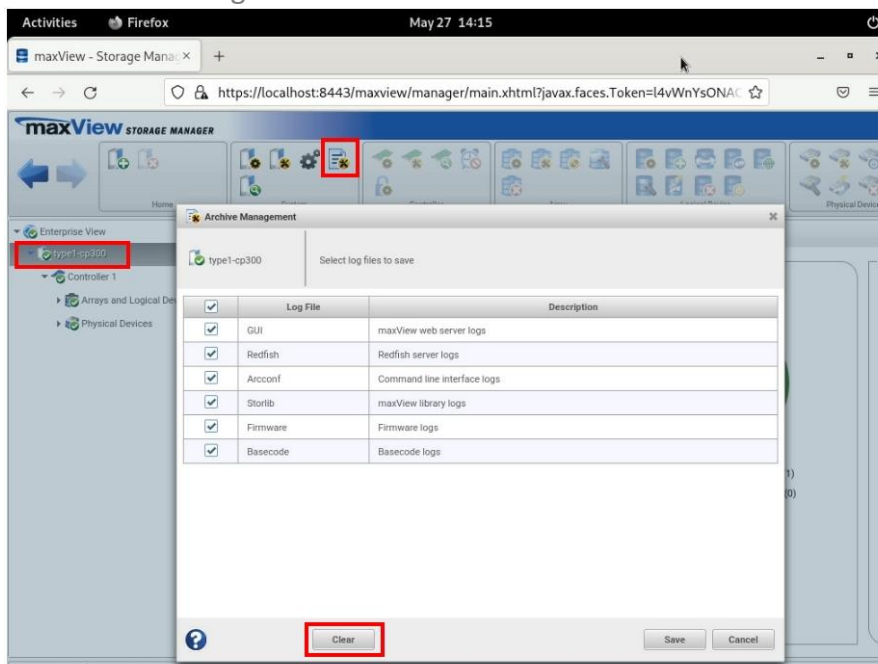
After Maxview is installed, goto <https://localhost:8443> via browser and enter the account / password.

Account: root

Password: siemens



Clear maxView log



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4.5 Execute CCTEST using Go2test

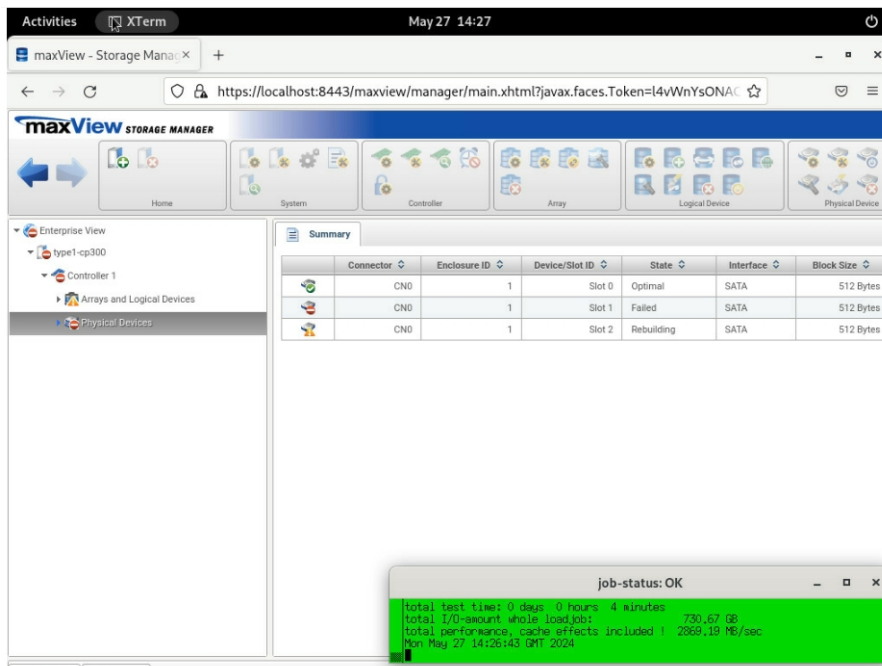
Test Case Selection (1/10): *

#	SERIAL Test Cases	SoE	DoP	TbS	User-Vars	Parameters	Select
01	16 - @TK - MaRCO-py3	☐		0h0m0s			
02	36 - @TK - MaRCO	☐		0h0m0s			
03	96 - @TK - PyRestart - Master	☐		0h0m0s		--testtime=6h --rscid=201800 --sccdisable=901	
04	96 - @TK - PyRestart - Master	☐		0h0m0s		--testtime=12h --rscid=380 --sccdisable=901	
05	96 - @TK - PyRestart - Master	☐		0h0m0s		--testtime=25 --rscid=1[5m] --load=SYSTEM --sccdisable=901	
06	96 - @TK - PyRestart - Master	☐		0h0m0s		--testtime=25 --rscid=1[5m] --load=HDD --sccdisable=901	
07	1001 - @TKLin - CCTEST-defaulttest	☐		0h0m0s		12:00:00	
08	2851 - @TKLin - LUGH	☐		0h0m0s		testtime=48:00:00 definitionfile=expert	
09	36 - @TK - MaRCO	☐		0h0m0s			
10	16 - @TK - MaRCO-py3	☐		0h0m0s			

(de)select all ☐ ☐ (de)select all

4.6 Start testing Hot spare

Remove one of the hard disks used to build the RAID and confirm that the hot spare HDD is automatically rebuilt.



The screenshot shows the maxView Storage Manager web interface. The 'Summary' tab displays the RAID status. The RAID is in a 'Rebuilding' state. A terminal window in the foreground shows the test results:

```

job-status: OK
Total test time: 0 days 0 hours 4 minutes
Total I/O-amount whole load job: 730,67 GB
Total performance, cache effects included: 2069,19 MB/sec
Mon May 27 14:26:43 GMT 2024
  
```

Check that IRM Web Mass Storage displays the rebuild message correctly.

Enclosures							
Status	Enclosure Number	Chain	Vendor	Product	Serial Number	Hardware Version	Logical ID
OK	378	-	FUJITSU	SAS4_32_Expander		0100	

Physical Disks										
Status	VMD	Enclosure	Port	Slot	Device Number	Interface Type	Type	Product	Physical Size [GB]	ID LED
Operational		378	0	0	8	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID
Failed Missing		378	0	1	9	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID
Rebuilding		378	0	2	10	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID

Logical Drives					
Status	Drive	Name	Logical Size [GB]	RAID Type	ID LED
Degraded	0	Logical Drive 1	190	RAID-1	

Create Logical Drive

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4.7 Check the LED

Check the correctness of the SUT HDD rebuild light signal and GLED light signal.
Please confirm whether the light signal during rebuild is correct according to the TPS of the current case



After the rebuild is completed, please insert an HDD of the same type and capacity as the HDD removed above into the SUT.

Physical Disks

	Status	VMD	Enclosure	Port	Slot	Device Number	Interface Type	Type	Product	Physical Size [GB]	ID LED
0	Operational		378	0	0	8	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID
0	Auto Replace Hot Spare		378	0	1	9	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID
0	Operational		378	0	2	10	SATA	HDD	ST2000NM000B-2TD100	1863.02	ID

Logical Drives

	Status	Drive	Name	Logical Size [GB]	RAID Type	ID LED
0	Operational	0	Logical Drive 1	100	RAID-1	

Create Logical Drive

4.8 Check all about hot spare log

After the test is completed, check the IRM Web SEL message for any abnormalities.

SEL								Clear Event Log	Save Event Log
	ID	Severity	Date	Code	Source	Description	Alert Group	CSS	
4		Info	2024-05-28 11:42:02	180010	SMS	HD on SAS controller 0 Enclosure 0 Slot 2 online	Disk Drivers & Controllers	—	
3		Info	2024-05-28 11:25:32	180013	SMS	HD on SAS controller 0 Enclosure 0 Slot 2 rebuilding	Disk Drivers & Controllers	—	
2		Critical	2024-05-28 11:25:32	180017	SMS	HD on SAS controller 0 Enclosure 0 Slot 1 dead	Disk Drivers & Controllers	—	
1		Info	2024-05-28 11:25:32	180018	SMS	HD on SAS controller 0 Enclosure 0 Slot 1 removed	Disk Drivers & Controllers	—	

4.9 Provide the logs for reference

Please attach CTEST, maxview log, SEL, SUT HDD rebuild GLED light signal... and other hot spare related logs in Testlogs folder for reference.

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5. 参考文件 Reference Document

TSR > TSR_HDD.xls